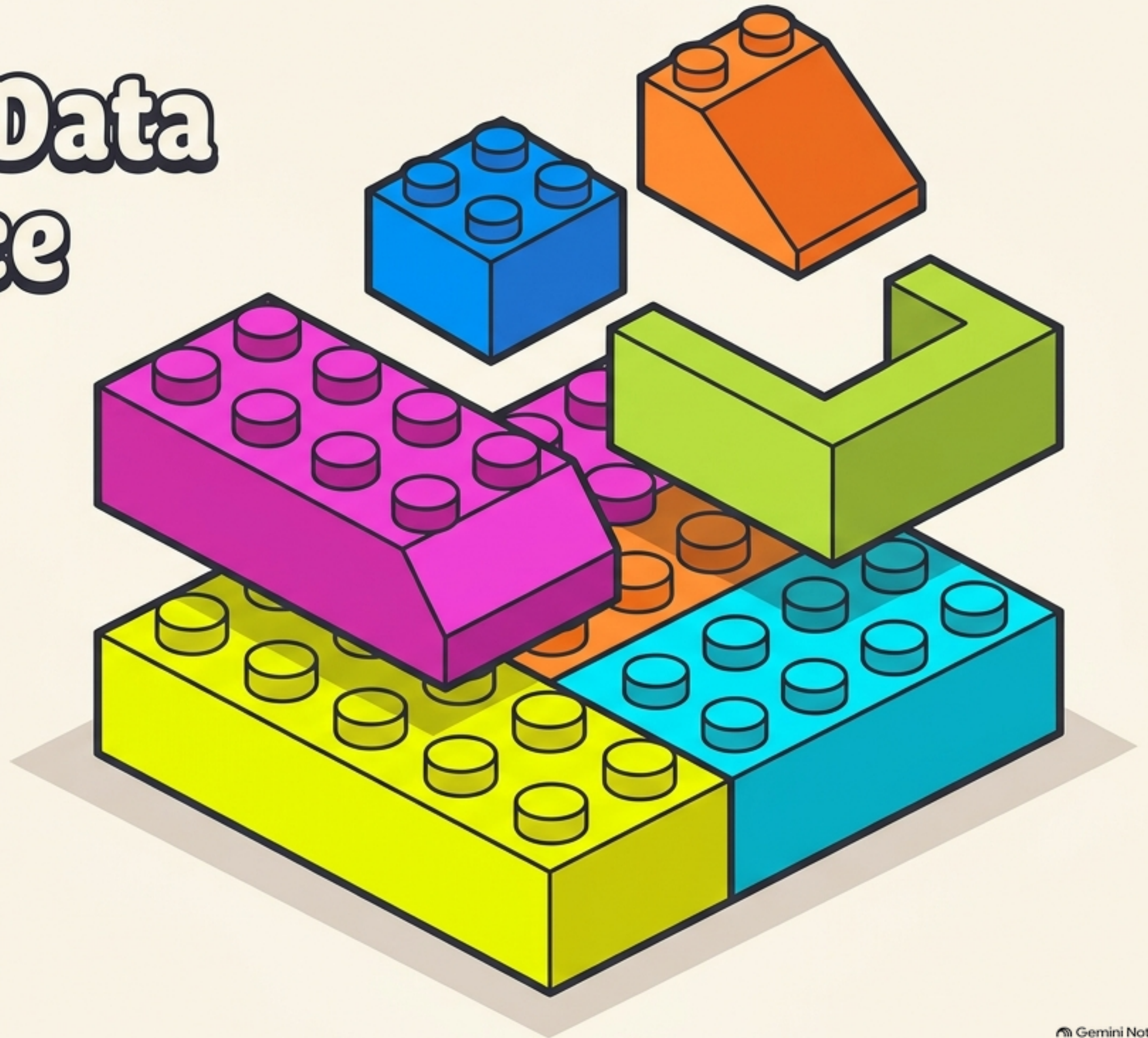


Building Big Data Piece by Piece

A visual, step-by-step guide
to constructing the Hadoop
ecosystem.

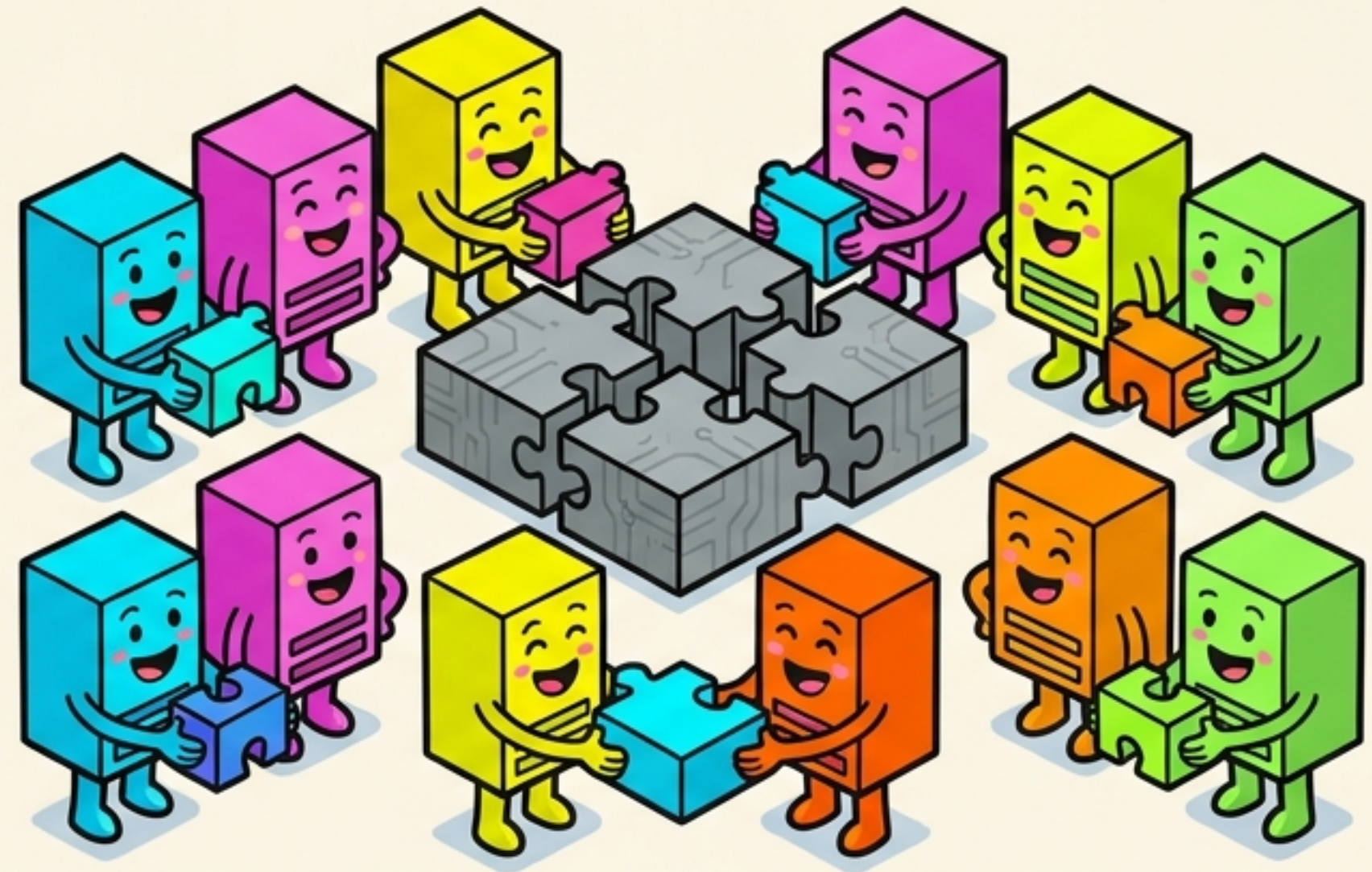


Traditional Storage Could Not Handle Growing Data



Standard File System

Controls how data is stored and read on a single hard disk. Eventually, the data gets too heavy for one machine.

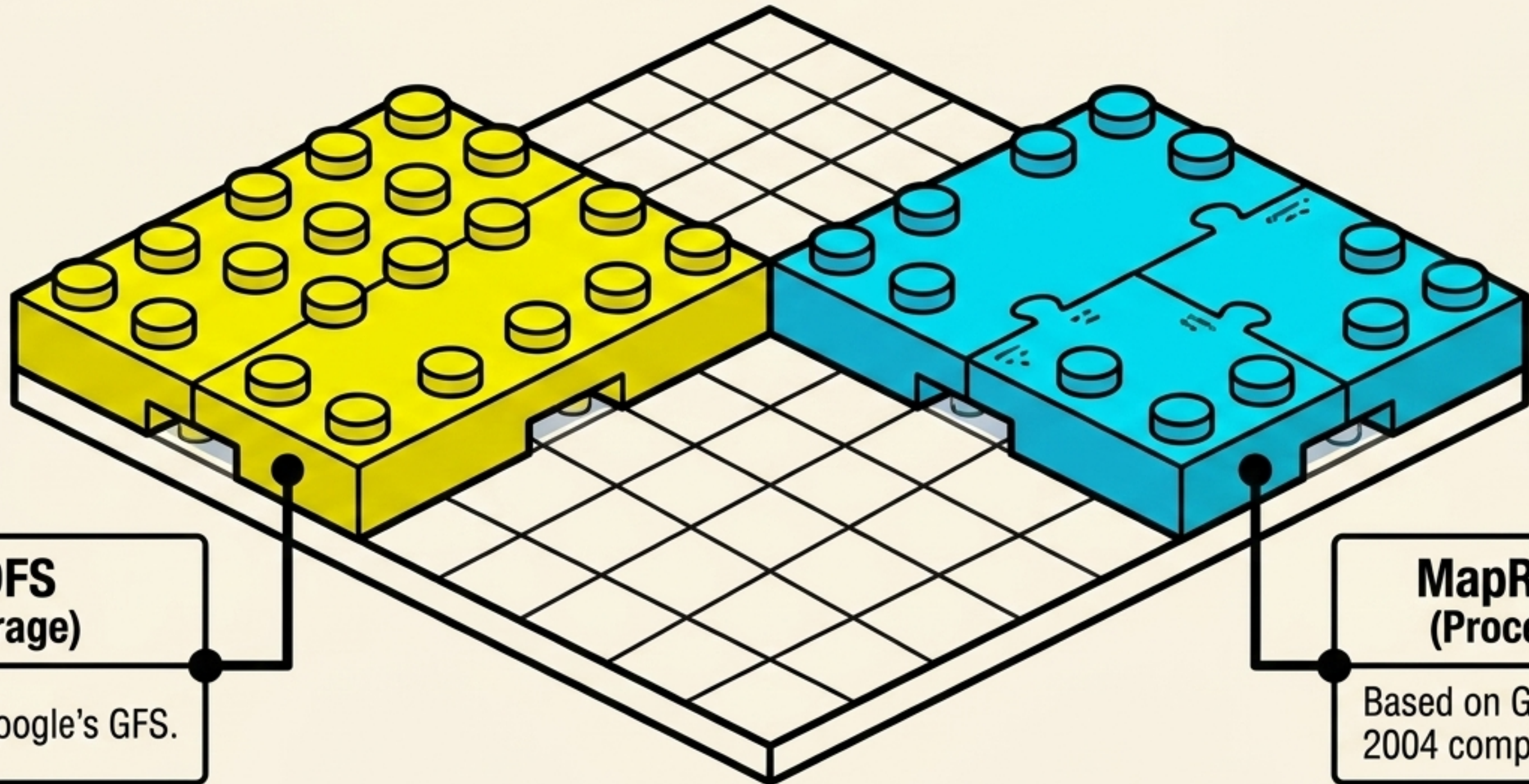


Distributed File System (Google GFS - 2002)

Spreads data across many machines. Instead of one giant computer, multiple smaller machines share the load.

The Foundation Started with Just Two Base Plates

In 2005, inspired by Google, Hadoop was created as a free, open-source project. If you downloaded it then, you only received these two foundational blocks:



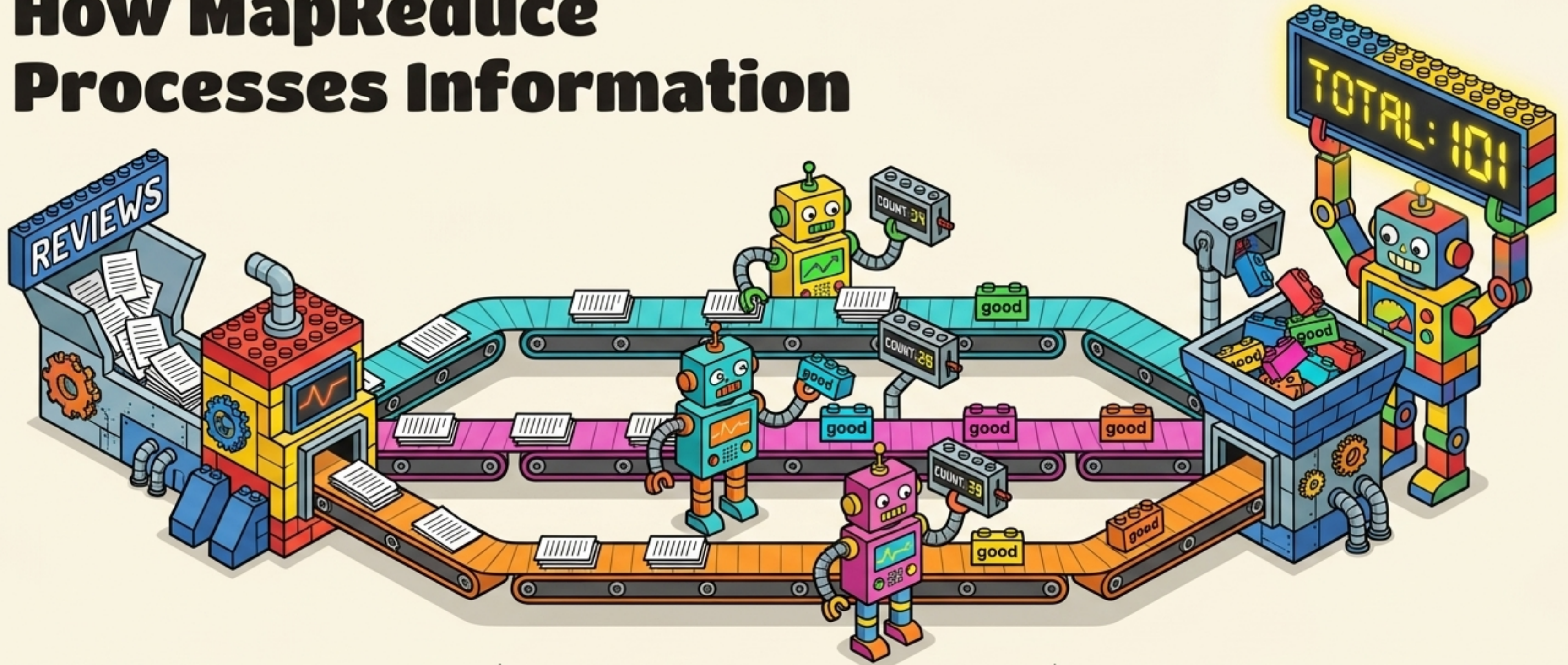
HDFS
(Storage)

Based on Google's GFS.

MapReduce
(Processing)

Based on Google's
2004 computing model.

How MapReduce Processes Information



Step 1: Divide

1,000 reviews are distributed across different machines.

Step 2: Map

Each machine independently counts the word "good" in its specific batch.

Step 3: Reduce

All the individual results are added together for the final count, making processing incredibly fast.

The Language Barrier Slowed Down Progress

Early Hadoop was powerful, but writing MapReduce programs required knowing Java.

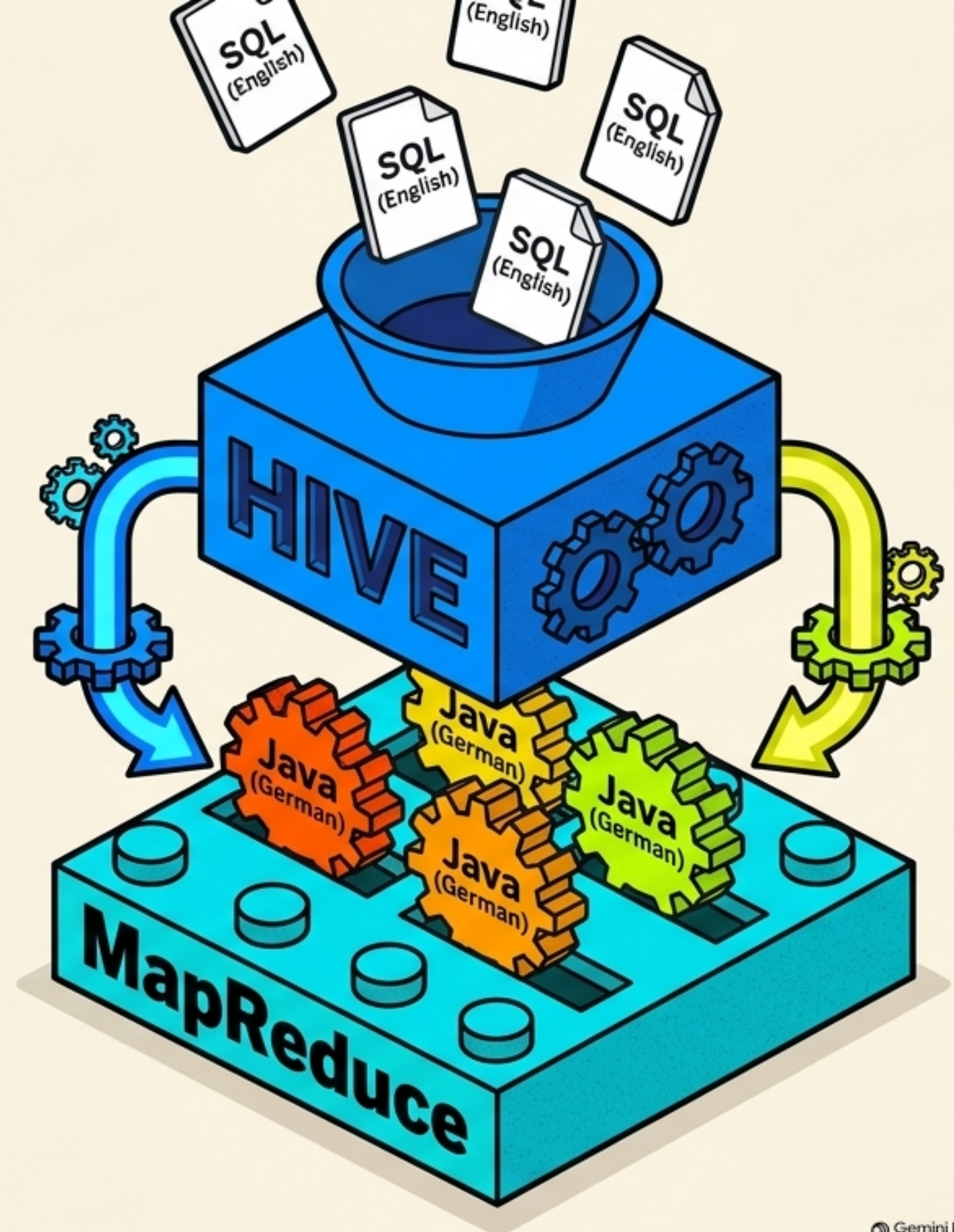
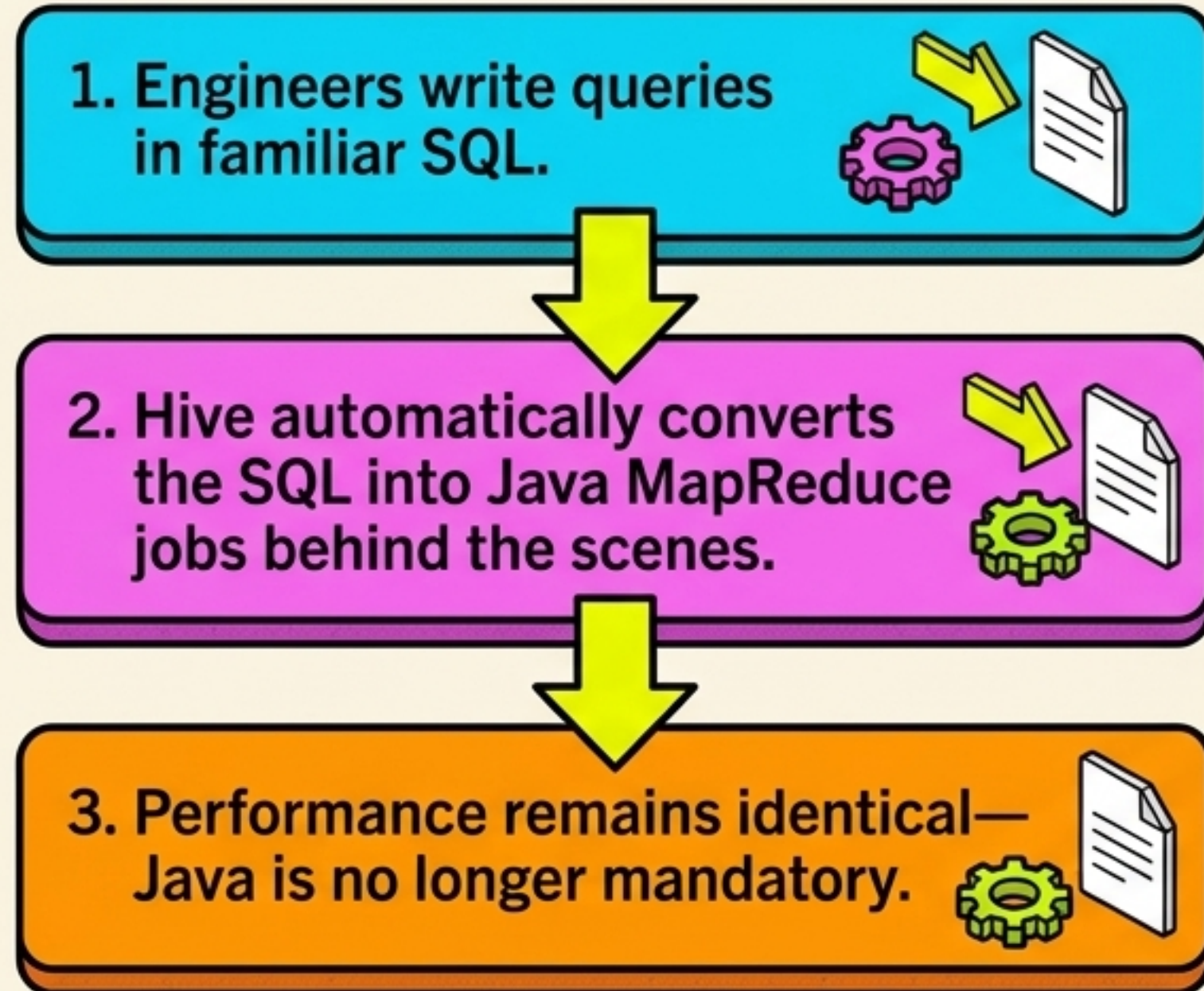


Imagine your entire team speaks **English** natively, but suddenly **all instructions must be written in German**. German works perfectly fine, but your team **struggles to build things quickly**.

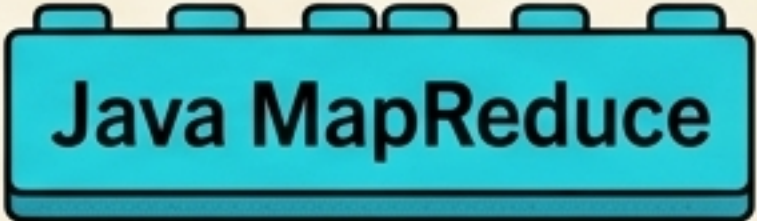
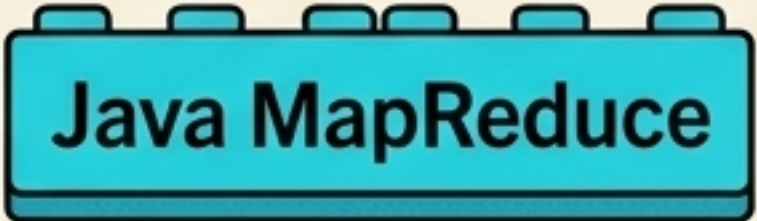
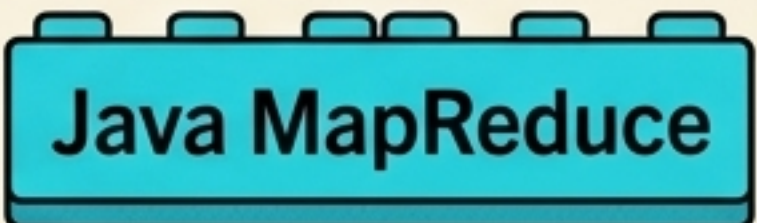
Facebook faced this exact issue: their engineers knew **SQL**, but writing in **Java** was **slowing** them down.

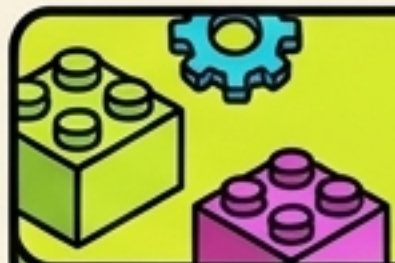
Hive Acts as the Universal Translator

Facebook engineered a solution by building a new tool directly on top of Hadoop.



Abstraction Blocks Hide the Complexity

Creator	Input Method	Primary Function	Output
Facebook 	SQL	Data Querying	
Yahoo 	Pig Latin Script	Data Scripting	
Yahoo 	Workflows	Job Scheduling	



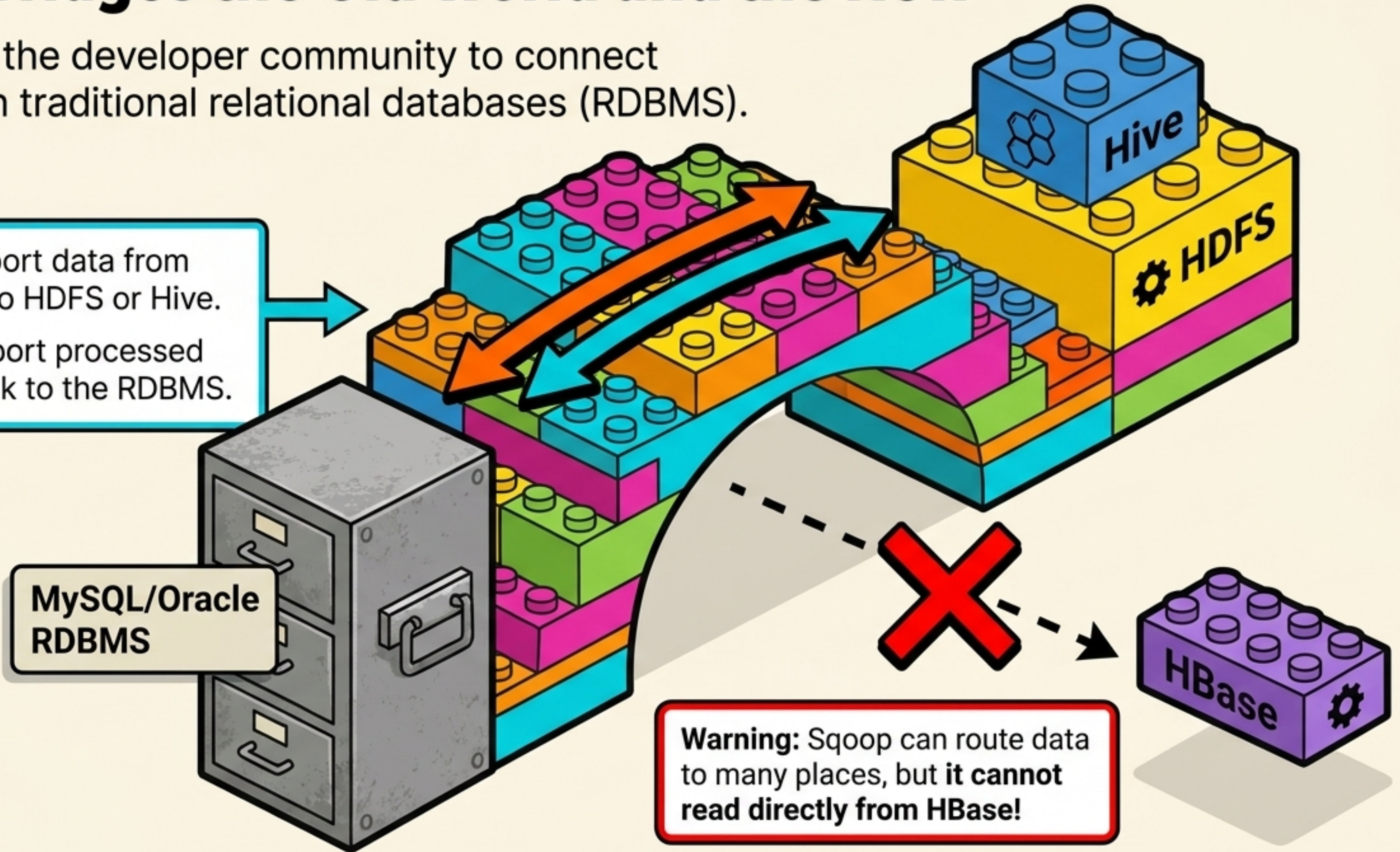
These tools are “abstractions.” They do not replace MapReduce; without the MapReduce base plate, they cannot function.



Sqoop Bridges the Old World and the New

Invented by the developer community to connect Hadoop with traditional relational databases (RDBMS).

- **Path 1:** Import data from RDBMS into HDFS or Hive.
- **Path 2:** Export processed results back to the RDBMS.



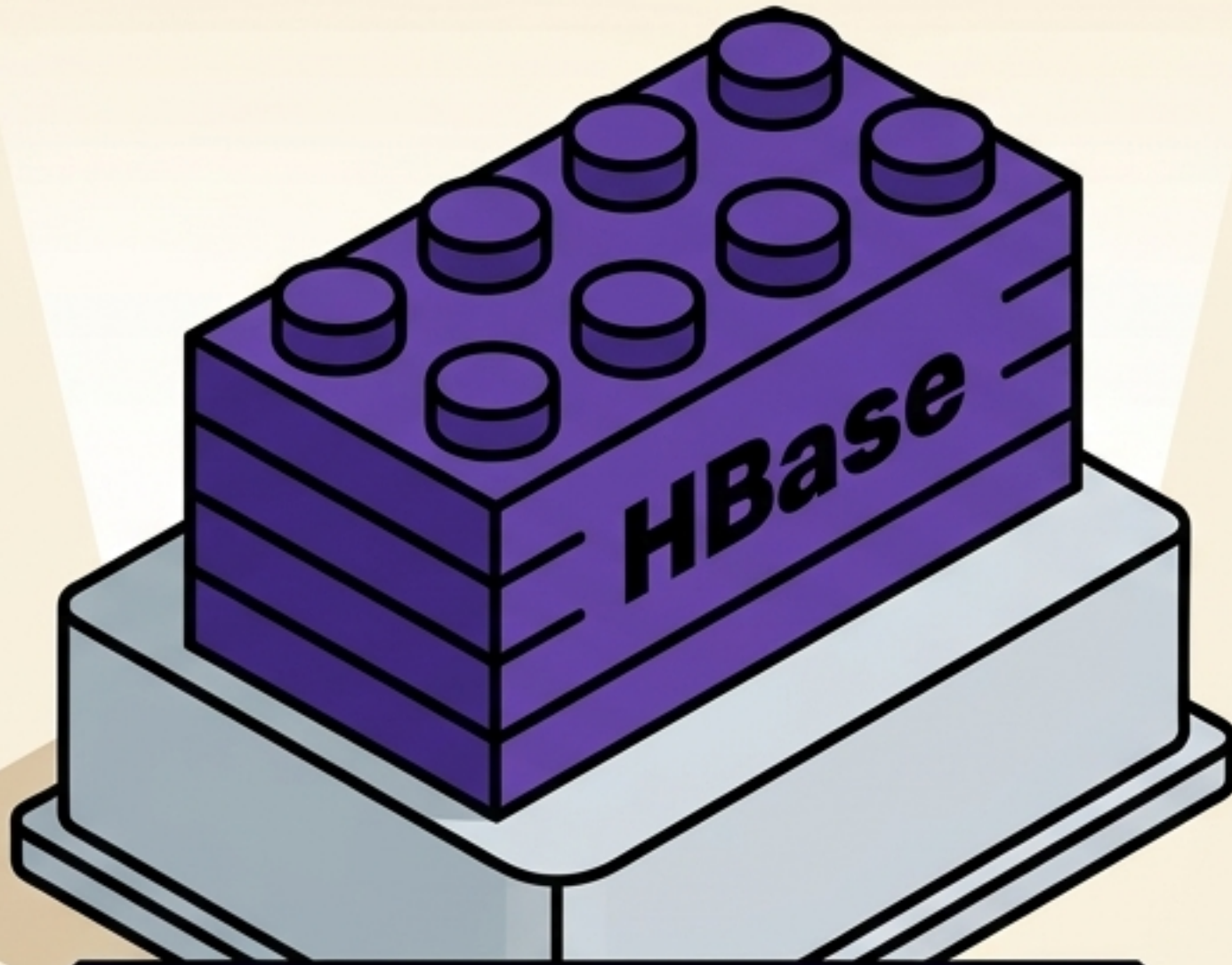
Flume Catches Real-Time Data Streams

Hadoop does not natively support real-time stream processing.

Flume acts as a messaging queue to solve this by bringing live data into Hadoop storage.



Specialized Blocks Complete the Set



HBase (Facebook)

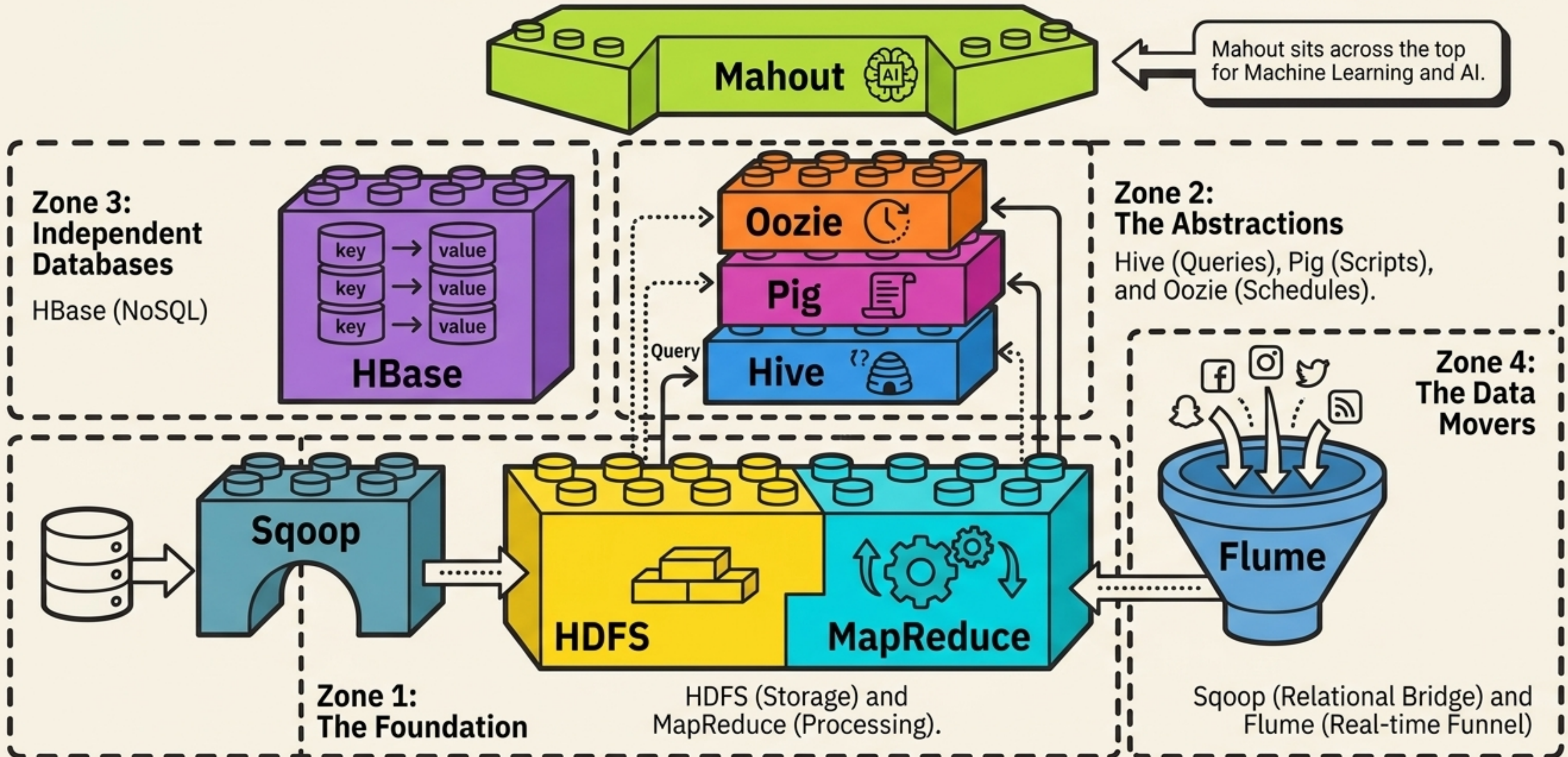
A big data NoSQL database. Handles massive, unconventional data that doesn't fit into tables. Built for incredibly fast random reads and writes.



Mahout

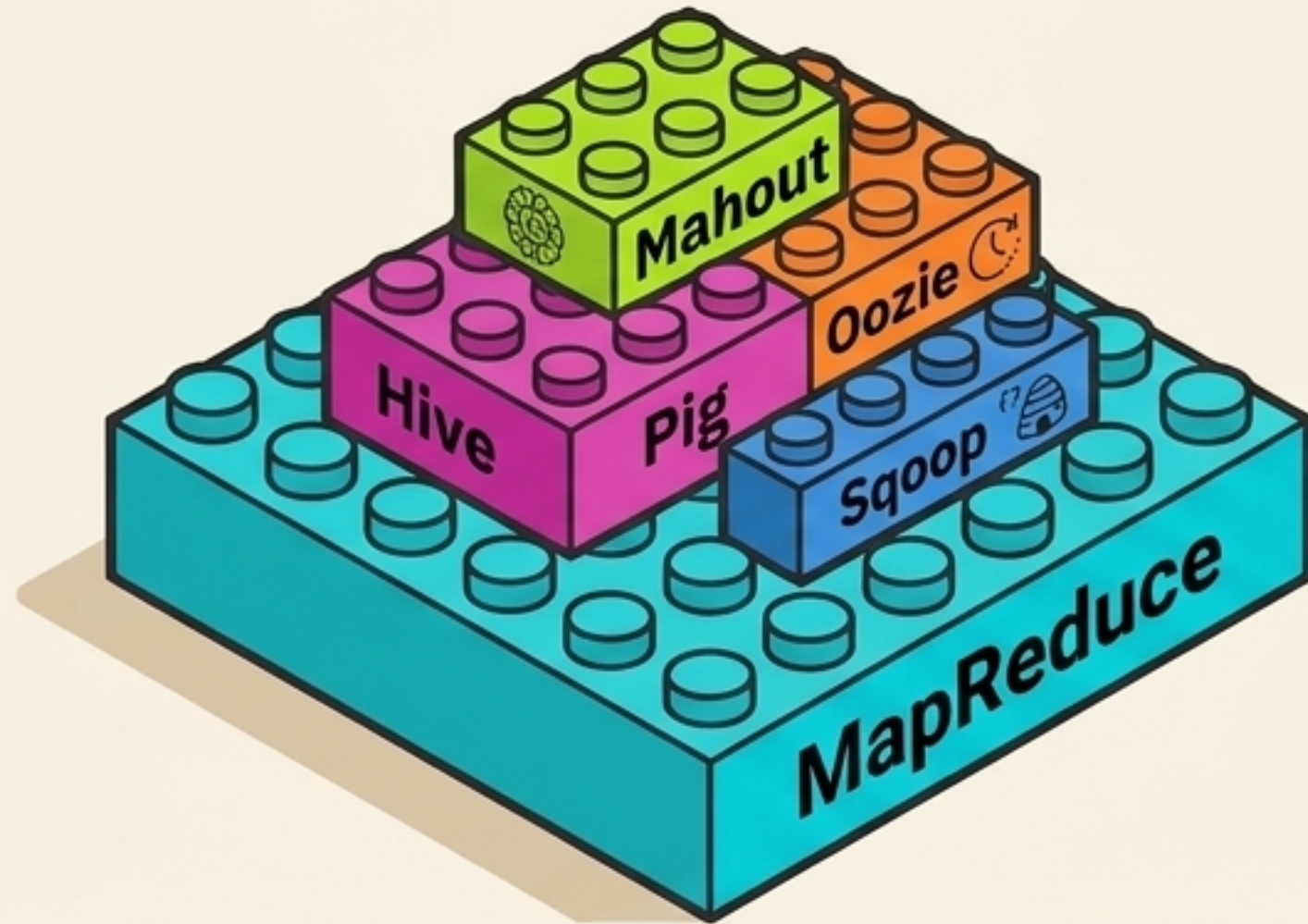
The machine learning block. Placed on top of Hadoop to simplify data science, AI, and advanced algorithms for developers.

The Expanded Hadoop Architecture



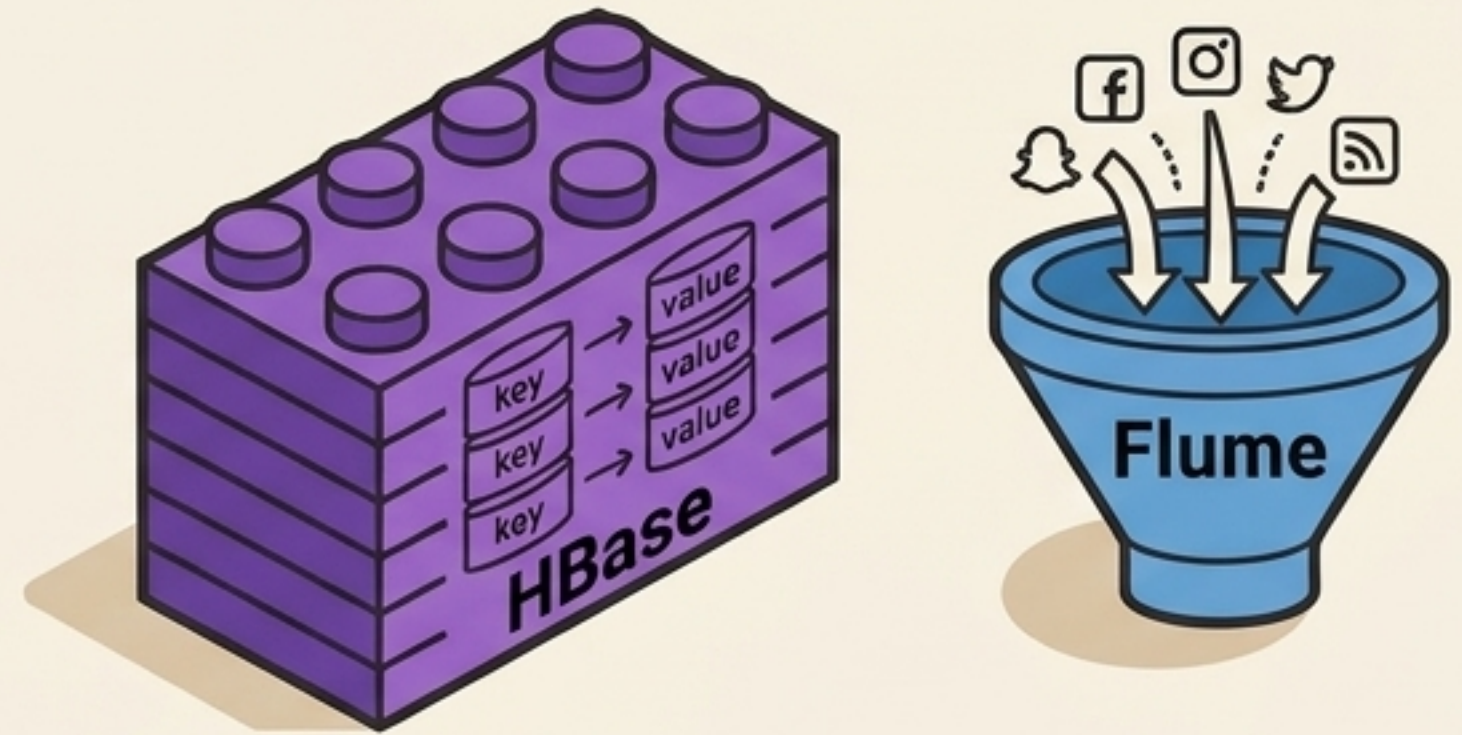
Sorting Blocks by their Processing Engine

Requires MapReduce



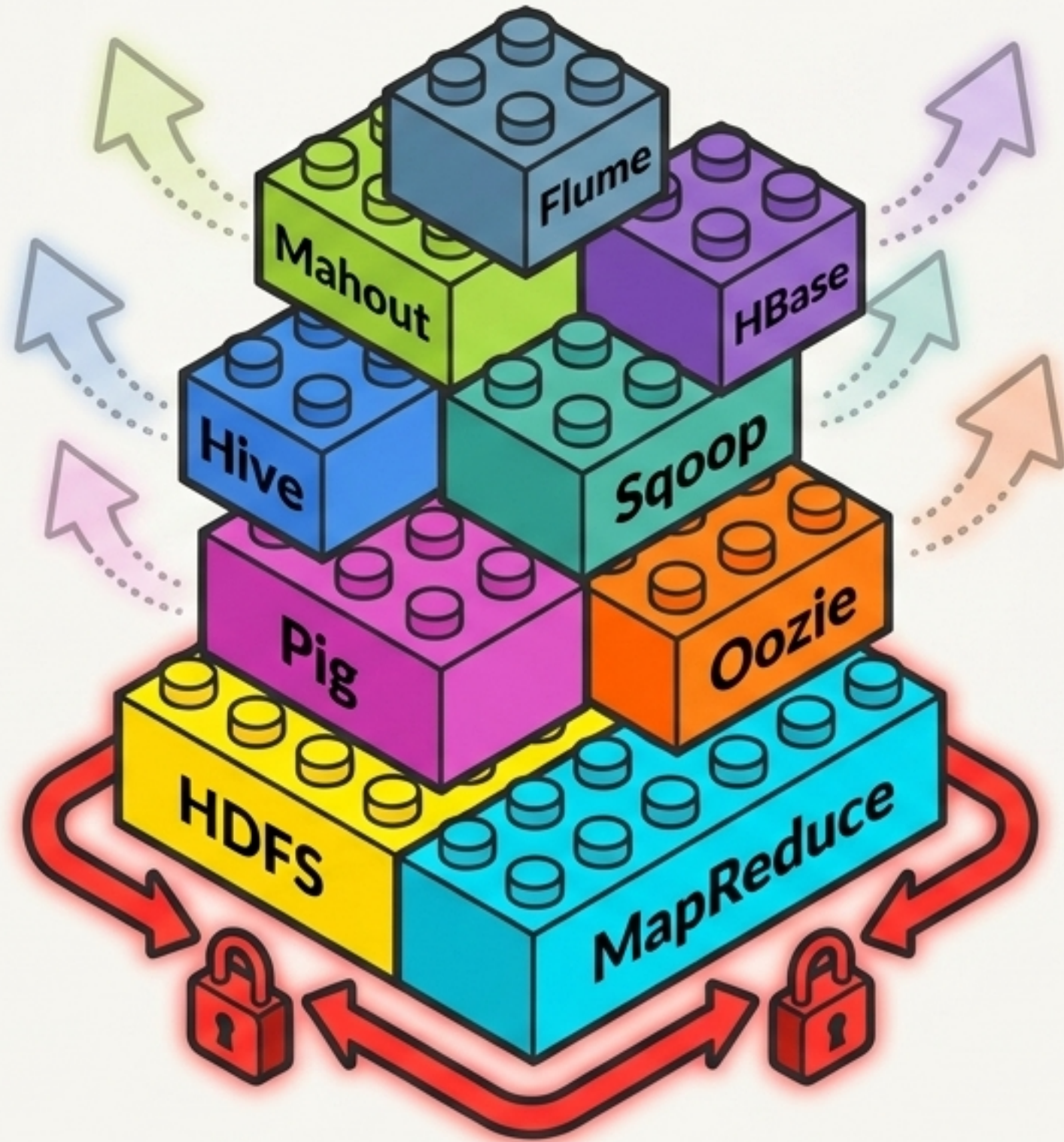
These tools sit on top of MapReduce and convert commands into Java jobs.

Independent Systems



These tools bypass MapReduce entirely to perform their specific storage or ingestion tasks.

The Strength of a Loosely Coupled Framework



Hadoop is highly customizable. You can snap on a new tool or remove an old one perfectly based on your exact requirements.

The Golden Rule of Hadoop

While the top ecosystem tools are completely optional, you cannot remove the **core base plates**. **HDFS** and **MapReduce** must remain for the structure to stand.